WORKSHEET 5.4

The First Derivative Test

Unit 5: Analysing Functions

Key ideas

- At a critical point c: if f' changes from **positive to negative**, f has a relative **maximum**; from **negative to positive**, a relative **minimum**.
- If f' does not change sign at c, there is no extremum there.
- The location is $x = c$; the extreme *value* is $f(c)$. Read which one the question wants.
- "$f'(c) = 0$" is not a justification. The sign change is.

Worked examples

Example 1 · A cubic

Find the relative extrema of $f(x) = x^3 - 6x^2 + 9x$.

Solution. $f'(x) = 3(x-1)(x-3)$, with signs $+, -, +$.

Relative maximum $f(1) = 4$; relative minimum $f(3) = 0$.

Example 2 · A repeated factor

A function has $f'(x) = (x+1)(x-2)^2(x-5)$. Classify its critical points.

Solution. Signs of f': $+$ for $x < -1$; $-$ on $(-1, 2)$ and on $(2, 5)$; $+$ for $x > 5$.

Maximum at $x = -1$, minimum at $x = 5$, neither at $x = 2$.

Example 3 · A cusp-type critical point

Classify the critical points of $f(x) = x - 3x^{1/3}$.

Solution. $f'(x) = 1 - x^{-2/3}$ is positive for $|x| > 1$ and negative for $0 < |x| < 1$.

Maximum at $x = -1$ (value 2), minimum at $x = 1$ (value -2), neither at $x = 0$.

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Practice

CORE

1. Find the relative minimum value of $f(x) = x^2 - 4x + 7$.

2. Find the relative maximum value of $f(x) = 6x - x^2$.

3. A function has f' changing from negative to positive at $x = 4$. What does f have there?

4. A function has $f'(x) = (x - 3)^2$. Does f have a relative extremum at $x = 3$?

EXAM LEVEL

5. Find and classify the relative extremum of $f(x) = x^4 - 4x$.

6. Find and classify the relative extremum of $f(x) = xe^{-x/3}$.

7. Find and classify the relative extremum of $f(x) = x^2 + \dfrac{2}{x}$ for $x > 0$.

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8. Find and classify the relative extrema of $f(x) = \sqrt{3}\sin x + \cos x$ on $(0, 2\pi)$.

Question 9 · Free-response practice

Let $f(x) = x^3 e^{-x}$.

(a) Find $f'(x)$ in factored form.

(b) Find the relative maximum value of f. Justify your answer.

(c) Explain why f has no relative extremum at $x = 0$.

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Worked solutions

1. $f'(x) = 2x - 4$ changes from negative to positive at $x = 2$: minimum value $f(2) = 3$.
2. $f'(x) = 6 - 2x$ changes from positive to negative at $x = 3$: maximum value $f(3) = 9$.
3. A relative minimum.
4. No: f' is positive on both sides of 3.
5. $f'(x) = 4x^3 - 4$ changes from negative to positive at $x = 1$: relative minimum, value -3.
6. $f'(x) = e^{-x/3}\left(1 - \dfrac{x}{3}\right)$ changes from positive to negative at $x = 3$: relative maximum, value $\dfrac{3}{e}$.
7. $f'(x) = 2x - \dfrac{2}{x^2}$ changes from negative to positive at $x = 1$: relative minimum, value 3.
8. $f'(x) = \sqrt{3}\cos x - \sin x = 0$ when $\tan x = \sqrt{3}$. Maximum at $x = \dfrac{\pi}{3}$ (value 2); minimum at $x = \dfrac{4\pi}{3}$ (value -2).
9. (a) $3x^2e^{-x} - x^3e^{-x} = x^2(3 - x)e^{-x}$.
   (b) f' changes from positive to negative at $x = 3$, so the relative maximum value is $f(3) = \dfrac{27}{e^3}$.
   (c) $f'(0) = 0$, but $x^2 \geq 0$ means f' is positive on both sides of 0: no sign change.

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